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Supplemental information

Deep molecular learning of transcriptional control of a synthetic CRE enhancer and its variants

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Supplementary information

Supplementary Figures 1-11

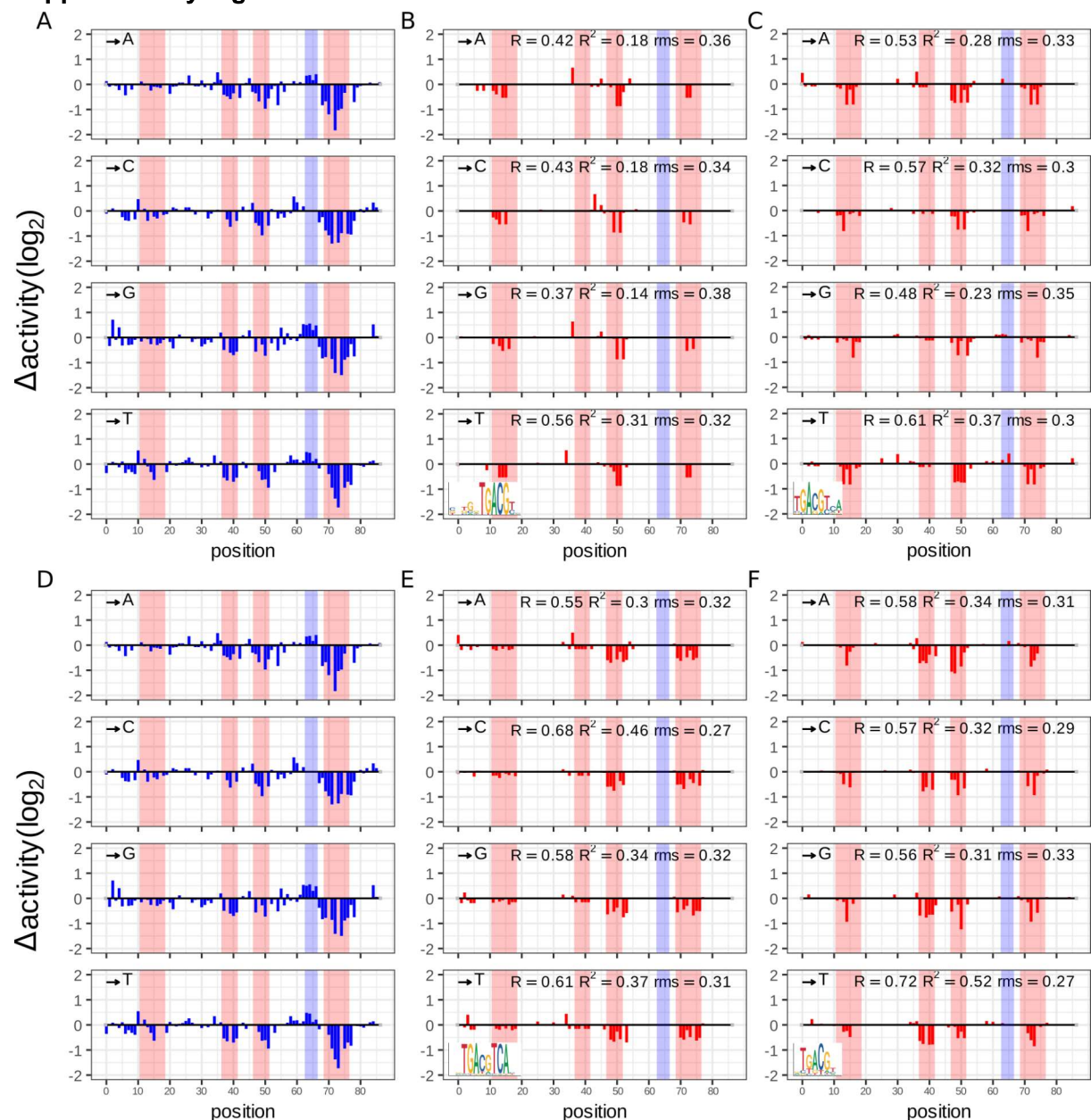
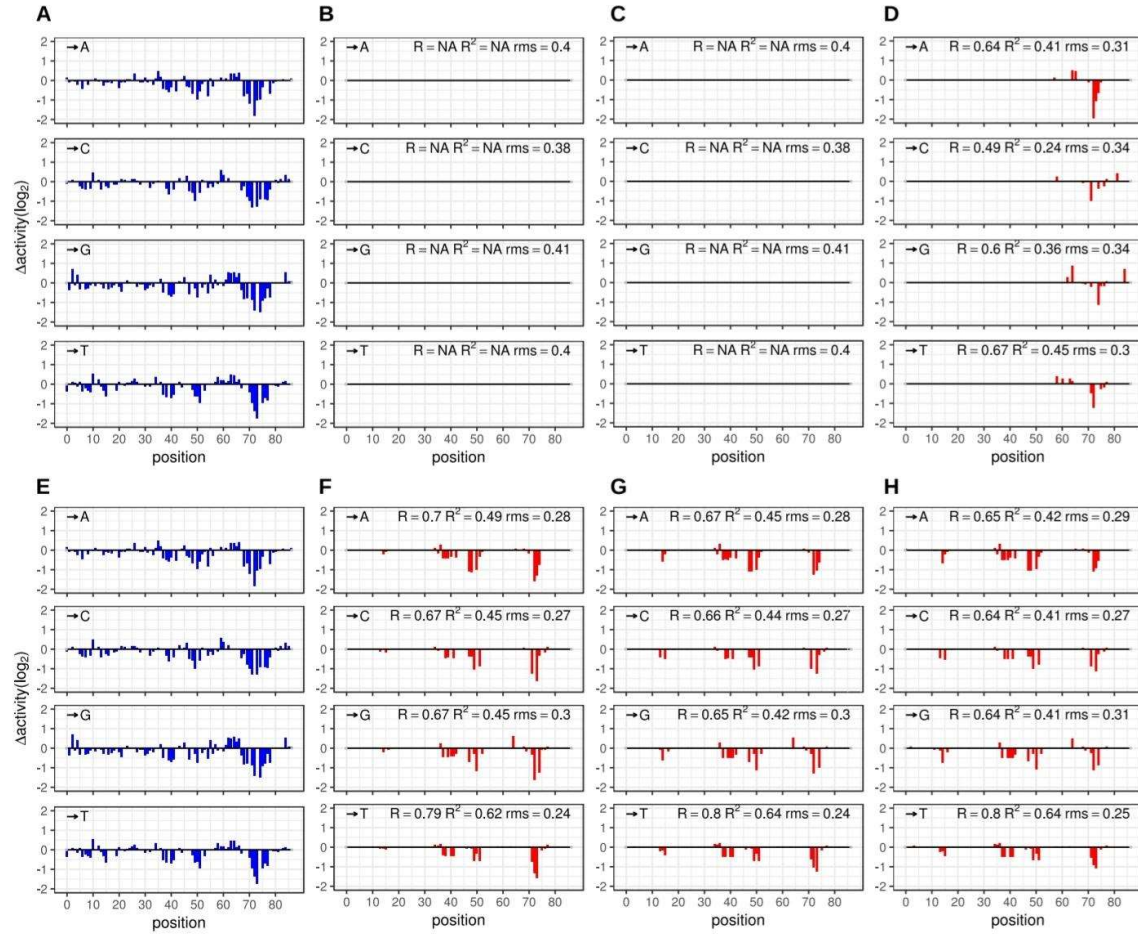


Figure S1. Baseline models with various CREB1 PWMs. Basal models that do not include any mechanism and TF other than the mentioned CREB1 PWM. (A,D) MPRA experimental data. (B) MA0018 1; SELEX PWM from JASPAR. (C) MA0018 2; compiled PWM from JASPAR. (E) MA0018 3; ChIP-seq PWM from JASPAR. (F) PBM PWM from DREAM5. Motif logos of PWM that were used in each model are shown at the left bottom of each panel.



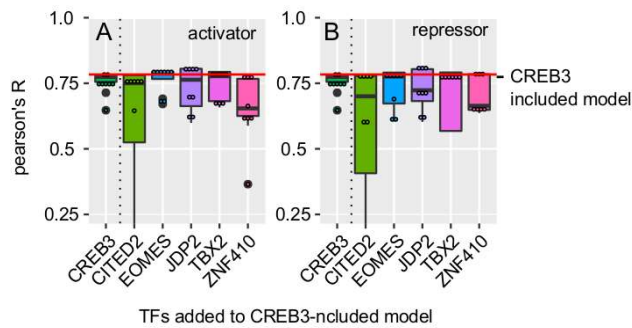


Figure S4. Addition of non-family TFs does not improve modeling results. Additional non-family TFs were added to CREB3-included model as (A) activators or (B) repressors. 10 models were trained for each group.

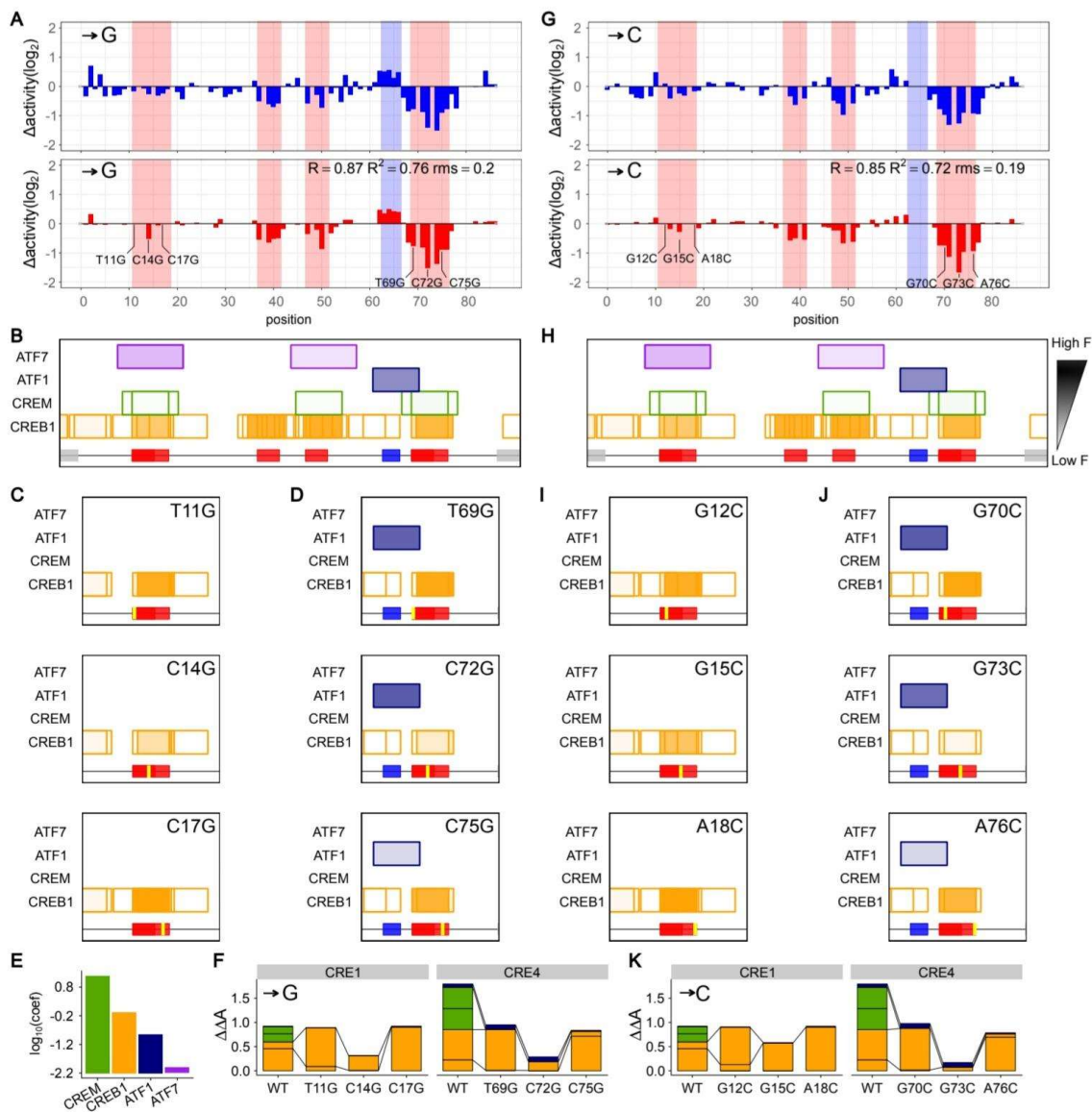


Figure S5. Functional binding site analysis of G/C substitutions. (A, G) Δ activity after G (A) or C (G) substitution. MPRA experimental result (top) and model calculation (bottom). (B) TFBSs in WT sequence. Each box represents TFBS, and transparency indicates fractional occupancy of the TFBS. TFBSs in CRE1 region (C, I) and CRE4 region (D, J) after variant introduction. Yellow box on the bottom represents the position of variant. (E) Activation coefficient of TFs. (F) Cumulative bar plot representing $\Delta\Delta A$ of CRE1/CRE4 binding TFs after variant introduction. (K) Cumulative bar plot representing $\Delta\Delta A$ of CRE1/CRE4 binding TFs after variant introduction.

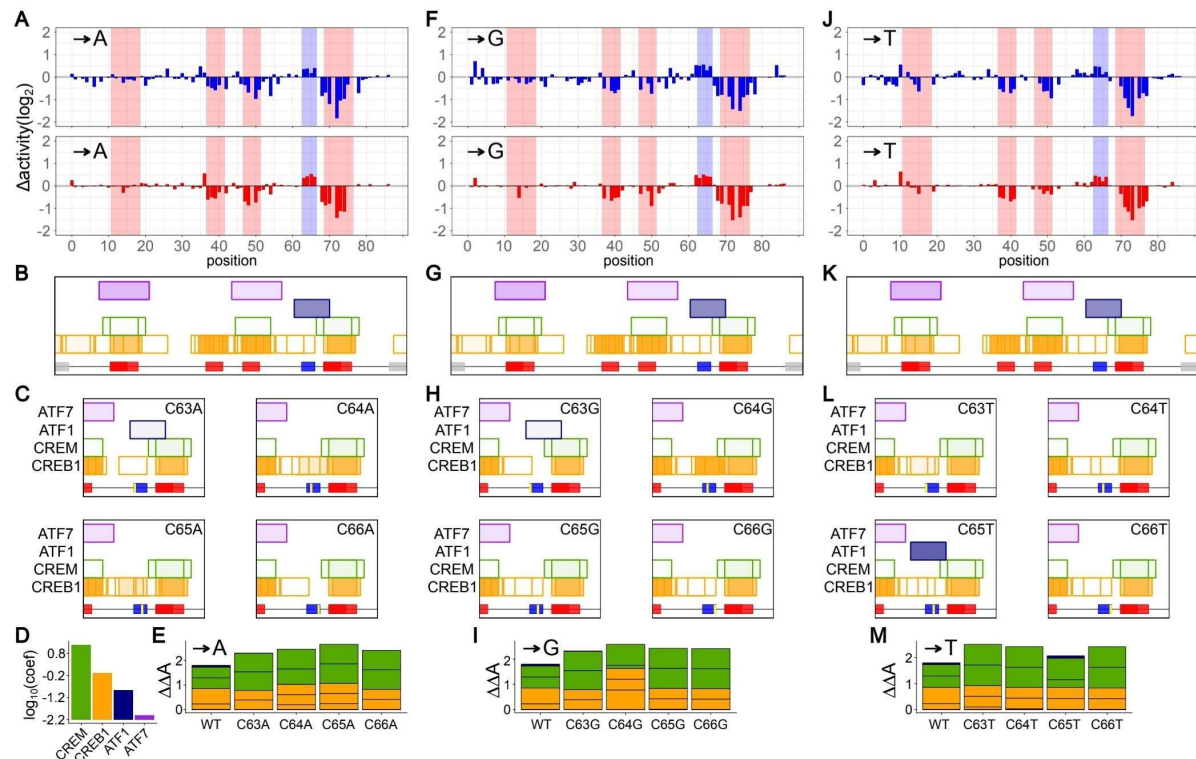


Figure S6. Functional binding site analysis in the cryptic region. (A, F, J) Δ activity after A, G, and T substitution. MPRA experimental result (top) and model calculation (bottom). (B, G, K) TFBSs in WT sequence (C, H, L) TFBSs in cryptic region after variant introduction. (D) Activation coefficient of TFs. (E, I, M) Cumulative bar plot representing $\Delta\Delta A$ of surrounding cryptic region binding TFs after variant introduction.

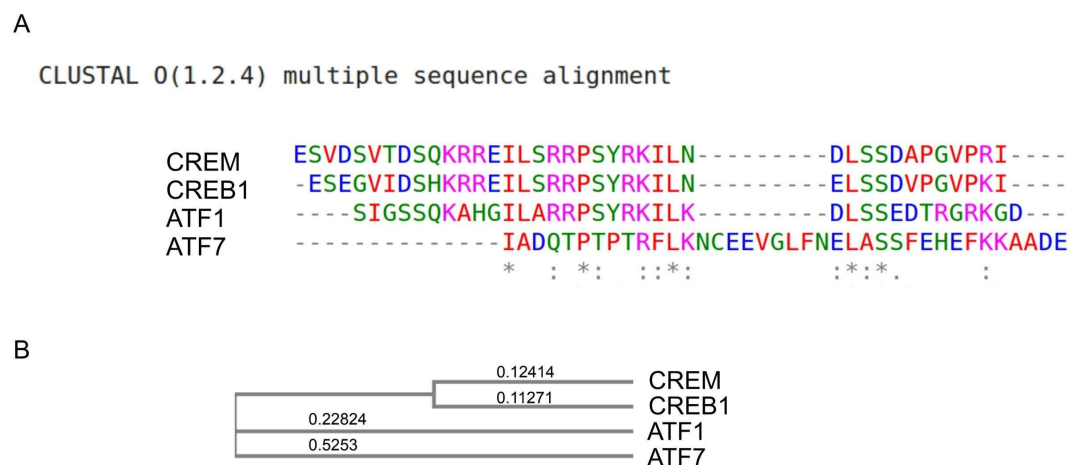


Figure S7. Comparison of TADs in ATF/CREB families. (A) Multiple sequence alignment of transcription activation domains (TADs) in ATF/CREB families. The color of each residue represents chemical properties. Red: small (small+hydrophobic [includes aromatic -Y], AVFPMILW); Blue: Acidic (DE); Magenta: Basic -H (RHK); Green: Hydroxyl, sulfhydryl, amine, G (STYHCNQG); Gray: Unusual amino/imino acid etc. Conserved residues are represented as asterisk(*). Colon(:) indicates conservation between groups of strongly similar properties. Period(.) indicates conservation between groups of weakly similar properties. (B) Phylogenetic tree from Clustal Omega.

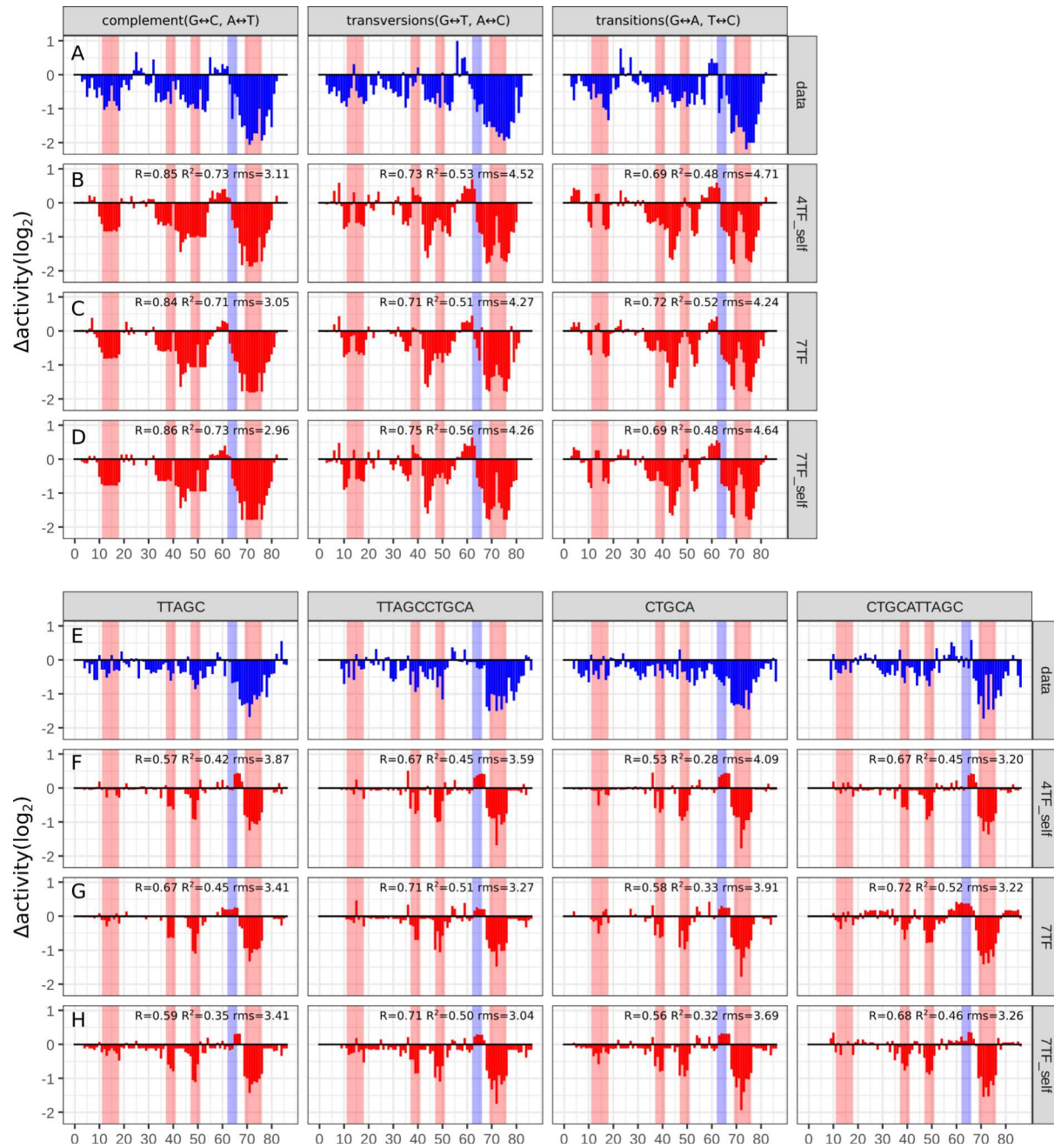


Figure S8. Gene expression changes caused by 8bp substitutions, 5bp or 10bp insertion.

Sequences with 8 bp substitutions (A~D) and 5 bp and 10 bp insertions (E~H) are shown at the top. Gene expression changes ($\Delta\text{activity}$) in (A, E) Experimental data, (B, F) 4TF model with self competition mechanism, (C, G) 7 ATF/CREB family TF model without self competition mechanism, (D, H) 7 ATF/CREB family TF model with self competition mechanism are shown for each position.